# Urban–Rural Gaps in Household Living Conditions in Nepal

Evidence from the 2022 Nepal Demographic and Health Survey

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**Abstract.** In 2022, 94 percent of rural Nepali households had electricity, while 21 percent used a clean main cooking fuel. Using 13,786 households from the 2022 Nepal Demographic and Health Survey, I compare nine living conditions across urban and rural municipalities. The largest urban advantages are in clean cooking fuel, finished floors, and finished walls. After adjusting for geography, household assets, and household-head characteristics, the gaps are 18, 15, and 13 percentage points. For housing materials and crowding, the estimated urban gap is 0.357 index points with an asset measure that excludes the outcomes. Adding the three building-material indicators to that measure lowers the estimate to 0.122. Nearly half of the households lacking clean cooking fuel, a finished floor, and finished walls live in urban municipalities. A rural-only rule would reach 51 percent of these households, while a rule based on missing at least one basic service would reach 41 percent.

**Keywords:** Nepal; household living conditions; urban–rural inequality; clean cooking; housing materials; wealth index; DHS.

**JEL codes:** I32, O18, R20.

**Data and code.** The 2022 NDHS household recode is available through The DHS Program following registration and approval. The microdata are not redistributed. The analysis was carried out in Python (pandas, statsmodels). Code, all aggregate results and a rendered run of the analysis are available at https://github.com/Porokmi/nepal-urban-rural-gaps-replication.

**Use of AI tools.** Claude (Anthropic) and ChatGPT (OpenAI) assisted with analysis code and checks of reported results.

## 1. Introduction

In Nepal's 2022 Demographic and Health Survey, 98 percent of the urban household population and 94 percent of the rural household population lived in homes with electricity. Clean cooking use was 52.6 percent in urban areas and 19.3 percent in rural areas (Ministry of Health and Population, New ERA, and ICF, 2023).[1] How large are the gaps in cooking, water, and housing conditions, and how much of each gap remains after accounting for household assets?

[1] The official NDHS benchmarks cited here describe the household population. Unless otherwise stated, estimates from this paper describe households and use DHS household sampling weights. Population-based comparisons additionally weight households by their number of usual residents.

I use the 2022 NDHS household sample to compare nine conditions across urban and rural areas. I group them into three counts covering basic services, cooking and water conditions, and housing materials and crowding. I then examine the indicators separately to see which conditions account for the differences in each count.

The NDHS final report reports service access and housing conditions separately for urban and rural areas (Ministry of Health and Population, New ERA, and ICF, 2023). I examine how these conditions occur together and how the urban–rural estimates change under different resource controls. Following Filmer and Pritchett (2001), I use principal components to construct an asset-based proxy for household resources. I select possessions and services that do not enter the outcome indices. The standard DHS wealth index includes housing and service characteristics, so it partly reflects the conditions I am comparing. I estimate models with both measures on the same sample within each outcome.

The closest precedent is Rheingans et al. (2014), who show that including water and sanitation items in a wealth index can overstate wealth disparities in sanitation access. Howe, Hargreaves, and Huttly (2008) find that, in their comparison, the choice of items had a greater effect on the resulting wealth index than the weighting method. This paper asks a related question about a different comparison: how including living-condition indicators in the wealth control changes the adjusted urban–rural gap. It shows which gaps remain after adjustment, which disadvantages occur together, and how much the estimated housing gap moves when those same conditions enter the wealth control.

Research in peri-urban Kathmandu Valley documents households using LPG and firewood together (Das et al., 2022). Acar, Kang, and Pokhrel (2026) also use the 2022 NDHS to examine how households choose fuels for cooking and heating. I compare the main cooking fuel across urban and rural households, so the indicator captures primary fuel use. The 2021 census housing report documents a large gap in cement flooring between urban and rural municipalities (National Statistics Office, 2024, Table 2.13). I use the NDHS to examine how much of the gap in finished floors remains after adjustment and how often households lack finished floors, finished walls, and clean cooking fuel together.

The largest unadjusted urban advantages are 36 percentage points for clean cooking fuel, 33 for finished floors, and 29 for finished walls. Adjusting for province, ecological zone, household assets, and household-head characteristics cuts those gaps roughly in half, to 18, 15, and 13 points. The adjusted basic-service count differs little by residence, even though urban households have greater electricity access and rural households are more likely to have basic sanitation.

The choice of wealth measure matters for housing materials and crowding. With geographic controls alone, the urban coefficient is 0.53 index points. Adding the DHS wealth index lowers it to 0.08, while adding the separate asset groups gives 0.36. The survey covers one year, so these estimates describe differences in 2022 and cannot show what caused them or how they changed over time.

The urban–rural gap also matters for how programs identify households in need. Rural households are more likely to lack clean cooking fuel, a finished floor, and finished walls together, but nearly half of households facing all three disadvantages live in urban municipalities. I compare how many of these households would be reached by rules based on residence, basic services, and asset or wealth groups.

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Differences between household and population percentages can therefore arise because household size varies with living conditions.

Sections 2–4 describe Nepal's context, the survey, and the measures. Sections 5 and 6 report the main estimates and show how individual gaps vary across provinces and ecological zones. Section 7 checks the sensitivity of the results, Section 8 examines which households different targeting rules would reach, and Section 9 discusses the implications and limitations. Appendix B reports a supplementary analysis of district migration prevalence.

# 2. Context

## 2.1 Geography and administrative residence

Nepal's Mountain, Hill, and Terai zones provide distinct geographic settings for household living conditions. The 2022 NDHS identifies these ecological zones alongside seven provinces and an administrative urban–rural classification. The survey uses the municipal classification updated in 2017, distinguishing 293 urban municipalities from 460 rural municipalities, or 753 local governments in total (Ministry of Health and Population, New ERA, and ICF, 2023).

Municipality boundaries encompass settlements of varying density. Using the Degree of Urbanisation method, an analysis of the 2021 census classified 27.07 percent of Nepal's population as urban, 39.75 percent as peri-urban, and 33.19 percent as rural (UN-Habitat, 2024). The comparisons in this paper follow the survey's municipal classification.

## 2.2 Basic services and cooking energy

The NDHS documents widespread electricity access in 2022: approximately 98 percent of the urban household population and 94 percent of the rural household population lived in households with electricity. Clean cooking use differed much more sharply, at 52.6 percent in urban areas and 19.3 percent in rural areas (Ministry of Health and Population, New ERA, and ICF, 2023). Electricity access alone therefore provides an incomplete description of household energy conditions.

Sanitation policy also provides relevant context. Nepal was declared open-defecation-free on September 30, 2019 (United Nations Sustainable Development Group, 2019). This declaration should be distinguished from the NDHS measure of basic sanitation, which requires an improved facility that is not shared with other households. The 2022 survey reports basic sanitation access among 72.6 percent of the household population. Likewise, an improved drinking-water source identifies a source category rather than establishing water safety or reliability.

Clean cooking was an explicit policy priority around the survey period. The December 2022 Country Action Plan for Transforming the Cookstoves and Fuels Market in Nepal describes reliance on biomass and imported LPG alongside plans to promote electric cooking. It identifies electricity infrastructure, appliance standards, consumer installment schemes, and access to credit as areas for action (Clean Cooking Alliance, 2022). These priorities provide context for examining cooking fuel separately from electricity access; the present analysis does not evaluate their implementation or effects.

## 2.3 Housing and the 2015 earthquake

The 25 April 2015 Gorkha earthquake destroyed more than half a million houses. Nepal's Post Disaster Needs Assessment designated 14 districts as crisis-hit and documented particular

vulnerability in poorer rural areas and in buildings lacking seismic-resistant features (National Planning Commission, 2015).

The 2022 survey records floor, wall, and roof materials. These observations provide no direct measure of earthquake resistance or reconstruction, and the analysis cannot separate rebuilt homes from other dwellings.

## 2.4 Migration and remittances

International migration is another important feature of Nepal's economic setting. The 2021 census recorded 2,190,592 absentees abroad, while World Development Indicators place personal remittances received at approximately 22.9 percent of GDP in 2022 (National Statistics Office, 2025; World Bank, World Development Indicators).

The census counts residents abroad, while the World Bank figure measures transfers received. Appendix B uses the census counts to compare districts. Households in a district receive the same migration value regardless of whether they receive remittances.

# 3. Data

## 3.1 Survey and sample

This study uses the household recode file of the 2022 Nepal Demographic and Health Survey (NDHS). New ERA conducted the survey under the direction of the Ministry of Health and Population, with technical assistance from ICF through The DHS Program. Fieldwork took place from January 5 to June 22, 2022 (Ministry of Health and Population, New ERA, and ICF, 2023).

The NDHS used a stratified, two-stage sampling design based on an updated frame from the 2011 Population and Housing Census. Each of Nepal's seven provinces was divided into urban and rural areas, creating 14 sampling strata. In the first stage, 476 primary sampling units were selected with probability proportional to size: 248 in urban areas and 228 in rural areas. Following household listing, the design called for selecting 30 households per cluster, giving a planned sample of 14,280 households. Interviews were completed in 13,786 households.

The household file contains observations from all 476 clusters, with between 10 and 32 completed household interviews per cluster and a median of 29. It covers all seven provinces, the three ecological zones, and 75 of Nepal's 77 districts. The unweighted sample includes 6,693 households in the Hill zone, 5,828 in the Terai, and 1,265 in the Mountain zone.

Urban and rural residence follows the administrative classification used by the survey, which distinguishes urban municipalities from rural municipalities under the classification updated in 2017. Urban residence therefore refers to administrative status rather than a classification based on population density or the extent of built-up land.

## 3.2 Weights and inference

Urban households account for 52.2 percent of completed interviews (7,195 households) and 67.1 percent of households after weighting (Figure 1). Unless otherwise stated, I weight each household

by hv005 divided by 1,000,000. The resulting percentages describe the share of households meeting each condition.

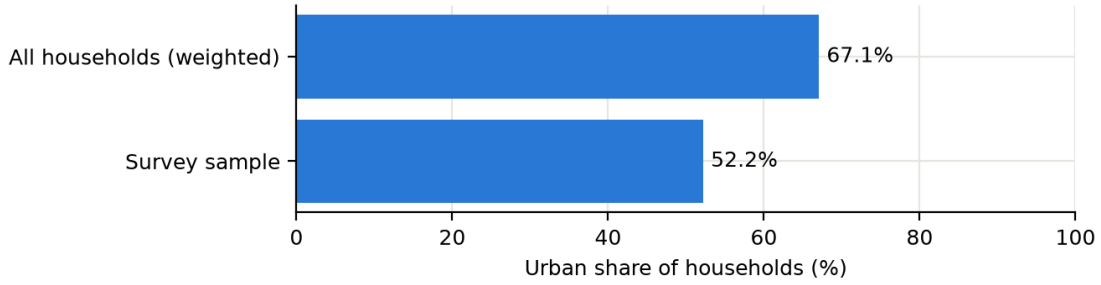



***Figure 1.*** *Urban share of the sample, unweighted and weighted*

I estimate the linear regressions by weighted least squares and cluster standard errors by primary sampling unit. The variance estimates account for clustering and omit the 14 sampling strata. Section 7.1 compares these standard errors with estimates that also account for the sampling strata.

Household percentages should be distinguished from the population percentages reported in the NDHS. For comparison with population-based benchmarks, household weights are multiplied by the number of de jure household members, or usual residents. The official report gives electricity access of approximately 98 percent among the urban household population and 94 percent among the rural household population. It reports basic sanitation access of 72.6 percent nationally (Ministry of Health and Population, New ERA, and ICF, 2023, Table 16.7).

The distinction between households and individuals is particularly relevant for cooking fuel. Clean cooking fuel is used by approximately 57.0 percent of urban households in this analysis, compared with 52.6 percent of the urban household population reported in the NDHS (Ministry of Health and Population, New ERA, and ICF, 2023, Table 2.4). The lower population percentage is consistent with clean-fuel use being less common among larger households.

## 3.3 Variables and classification

The analysis examines nine indicators of household living conditions: electricity access; an improved drinking-water source; basic sanitation; clean cooking fuel; water on the premises; finished floor, wall, and roof materials; and adequate sleeping space. These are constructed from electricity access (`hv206`), drinking-water source (`hv201`), sanitation facility type and sharing status (`hv205`, `hv225`), main cooking fuel (`hv226`), water collection time (`hv204`), building materials (`hv213`–`hv215`), and household size relative to sleeping rooms (`hv009`, `hv216`). Section 4 explains how the indicators are combined into summary indices.

Response categories are classified using the WHO/UNICEF Joint Monitoring Programme definitions for drinking water and sanitation, the WHO classification of clean cooking fuels, and DHS categories for finished building materials. An improved drinking-water source describes the type of source; it does not establish that the water is safe or consistently available. Basic sanitation requires an improved facility that is not shared with other households. Biogas-attached toilets are classified as improved, following the Nepal report, and flush toilets with an unknown destination are classified as

improved, following DHS/JMP guidance.[2] Appendix Table A1 provides the complete classification and the number of households in each response category.

Unclassified "other" responses are treated as missing rather than automatically counted as deprivation. Cooking-fuel status is also treated as missing for the 40 households that report no cooking at home. For sanitation, missing sharing information does not change the score of a household with an unimproved facility or no facility, since neither meets the definition of basic sanitation. An improved facility with unknown sharing status would be treated as missing. No individual indicator is missing for more than 40 households, or approximately 0.3 percent of the sample.

## 3.4 Household resources and controls

The standard DHS wealth index includes housing materials, drinking-water sources, and sanitation characteristics that overlap with the outcomes examined here. I therefore construct an alternative resource measure from eleven household possessions and services that do not directly enter the outcome indices: radio, television, refrigerator, bicycle or rickshaw, motorcycle or scooter, car/truck/tractor, landline telephone, mobile telephone, watch, computer, and bank account.

All eleven items are observed for every household. Following the PCA-based approach of Filmer and Pritchett (2001), the asset score is the first principal component of the standardized indicators. It serves as a proxy for household economic resources rather than a direct measure of income or expenditure.

Households are assigned to five groups using the weighted distribution of asset scores. Identical scores are kept together, so group sizes range from 14.8 to 23.8 percent of households rather than exactly 20 percent. The continuous asset score is also used as an alternative specification.

Additional controls include province, ecological zone, and the household head's education, sex, and age. Head education is obtained from the household roster and is missing for 25 households. To keep the main specifications comparable, these households are excluded from all models in the specification ladder, including those without head controls. The estimation sample is held constant across specifications within each outcome.

## 3.5 District migration data

The supplementary analysis uses district absentee-abroad counts from Annex 5 of *Internal Migration in Nepal*, Thematic Report V of the 2021 National Population and Housing Census (National Statistics Office, 2025). The transcribed district counts sum to the published national total of 2,190,592 absentees abroad and satisfy the table's in-, out-, and net-migration arithmetic. These provide checks on transcription consistency.

For each district, absentee-abroad counts are divided by the table's native population, excluding institutional households, and expressed per 100 native residents. The resulting measure ranges from

[2] The NDHS report appears to classify flush toilets with an unknown destination as unimproved: doing so reproduces its basic-sanitation figure of 72.6 percent when rounded to one decimal place, while this analysis's classification gives 72.66 percent. The category covers 11 households and changes the estimate by about 0.05 percentage points.

approximately 1.0 to 19.5. All 75 districts represented in the DHS household sample are matched. Manang and Rukum East are not represented in that sample.

The district absentee rate assigns a common value to all households in a district. The data used here contain no measure of which households receive remittances or how much they receive.

# 4. Measurement

## 4.1 Three dimensions of household living conditions

I organize the nine household indicators into three dimensions: basic services, cooking and water conditions, and housing materials and crowding. Each indicator takes a value of one when the specified condition is met and zero otherwise. An index is treated as missing if any of its components is missing.

The **Basic Services Index (BSI)** ranges from zero to three and adds electricity access, an improved drinking-water source, and basic sanitation:

$$BSI_i = Electricity_i + ImprovedWater_i + BasicSanitation_i.$$

Basic sanitation requires an improved facility that is not shared with other households. The index measures access to these services rather than their reliability or quality. Electricity access does not establish a continuous supply, and an improved drinking-water source does not establish that the water is safe.

The **Cooking and Water Conditions Index (CWI)** ranges from zero to two:

$$CWI_i = CleanFuel_i + WaterOnPremises_i.$$

Clean cooking fuels include electricity, solar energy, LPG, natural gas, and biogas. Water on the premises is identified from the reported water-collection-time variable. These indicators describe conditions relevant to cooking and water collection, including the effort required to meet everyday household needs. The fuel indicator records the main fuel and does not capture the use of additional fuels or direct exposure to indoor air pollution.

The **Housing Materials and Crowding Index (HMCI)** ranges from zero to four:

$$HMCI_i = FinishedFloor_i + FinishedWalls_i + FinishedRoof_i + NotOvercrowded_i.$$

Building materials follow the DHS finished-material categories. A household is classified as not overcrowded when it has no more than three household members per room used for sleeping. This cutoff is an operational choice, not a complete measure of housing adequacy. Similarly, finished materials do not directly establish structural safety, maintenance quality, insulation, or earthquake resistance.

## 4.2 Interpretation and weighting

Each condition contributes one point to its index. A one-point difference means one additional condition is met. The weights reflect the choice to report a count, with no valuation of individual conditions.

A CWI score of one can describe a household with clean fuel that collects water away from home or a household with water on the premises that cooks with polluting fuel. The individual indicators distinguish those households.

The correlation evidence in Figure 2 reinforces the need to retain this detail. The correlation between clean cooking fuel and water on the premises is approximately 0.17, while finished floors and walls are much more closely related, with a correlation of approximately 0.74. At the index level, BSI correlates approximately 0.19 with CWI and 0.22 with HMCI; CWI and HMCI correlate approximately 0.52.

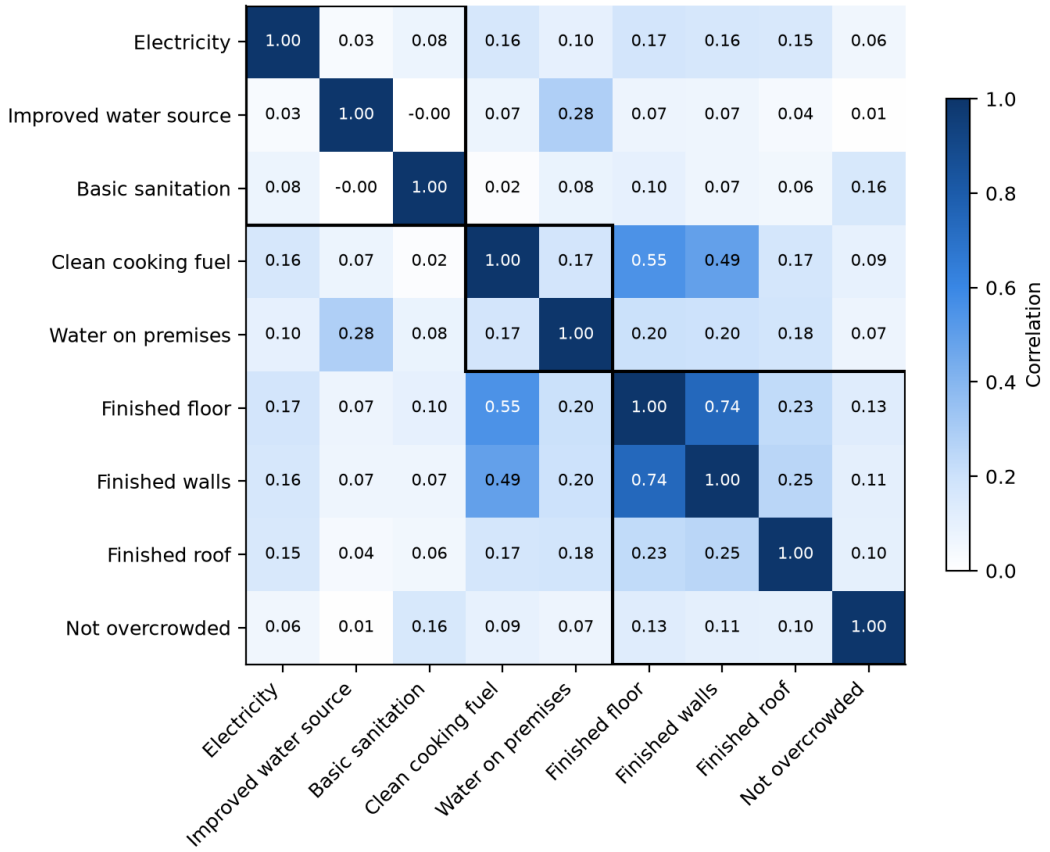



***Figure 2.*** *Correlations between the nine components (boxes mark the three indices)*

Electricity and improved-water access vary little across the sample, which may help explain the weak correlations involving basic services. The indices group conditions by subject, and the component estimates show how each condition differs by residence.

## 4.3 Household profiles

To describe how these conditions occur together, I assign households to four mutually exclusive profiles:

| Profile | Definition |
|---|---|
| **Basic Services and Living Conditions Met** | BSI = 3, CWI = 2, and HMCI ≥ 3 |
| **Basic Services Only** | BSI = 3 and CWI = 0 |

| Profile | Definition |
|---|---|
| **Limited Basic Services** | BSI ≤ 1 |
| **Intermediate** | All other households with complete index information |

I use the profiles to describe which conditions households meet together. They are an analytical typology based on chosen cut-offs, not official poverty categories or validated stages of development. Households with missing information in any index are excluded from profile comparisons.

The labels should be read alongside their definitions. “Basic Services and Living Conditions Met” requires all three basic services, both cooking and water conditions, and at least three of the four housing conditions; it does not require all four HMCI components. “Basic Services Only” identifies households with all three basic services but neither CWI condition, regardless of their housing materials. “Limited Basic Services” identifies households meeting at most one basic-service condition; it does not measure deprivation in cooking, water, or housing conditions.

The profiles complement the index and component results by showing whether basic services coexist with other favorable conditions. Comparisons across residence and asset groups describe the distribution of these combinations without implying that households necessarily move through the profiles in a fixed sequence.

## 4.4 Alternative asset measure

I standardize the eleven indicators listed in Section 3.4 and take their first principal component as the asset score (Filmer and Pritchett, 2001). I orient the score so that higher values indicate greater asset ownership.

For the grouped specification, households are assigned to five groups using the weighted score distribution. Identical scores remain together, producing groups of unequal size. I also estimate a specification using the continuous score and examine an alternative score constructed with survey-weighted PCA.

Asset ownership can reflect household resources, local infrastructure, and preferences. Where households live can also shape which assets they acquire through differences in employment and access to credit. The adjusted gaps describe differences after accounting for the observed asset measure and other included characteristics.

# 5. Results

## 5.1 Differences in individual household conditions

In Table 1 and Figure 3, urban households lead by 4.0 percentage points in electricity access and 2.4 points in improved drinking-water sources. For basic sanitation, the direction reverses. Coverage is about 70 percent in both areas, with a 1.5-point rural advantage. Facility sharing is more common in urban areas (Section 6).

***Table 1.*** *Households meeting each condition (weighted %, NDHS 2022). Gap = urban minus rural, percentage points.*

| Condition | All | Urban | Rural | Gap (pts) |
|---|---|---|---|---|
| Electricity | 96.4 | 97.7 | 93.7 | 4.0 |

| Condition | All | Urban | Rural | Gap (pts) |
|---|---|---|---|---|
| Improved water source | 98.0 | 98.8 | 96.4 | 2.4 |
| Basic sanitation | 69.4 | 68.9 | 70.4 | -1.5 |
| Clean cooking fuel | 45.2 | 57.0 | 21.2 | 35.7 |
| Water on premises | 85.1 | 88.1 | 78.9 | 9.2 |
| Finished floor | 53.4 | 64.1 | 31.5 | 32.6 |
| Finished walls | 60.1 | 69.5 | 40.8 | 28.7 |
| Finished roof | 95.2 | 97.4 | 90.8 | 6.5 |
| Not overcrowded | 87.1 | 88.0 | 85.3 | 2.7 |

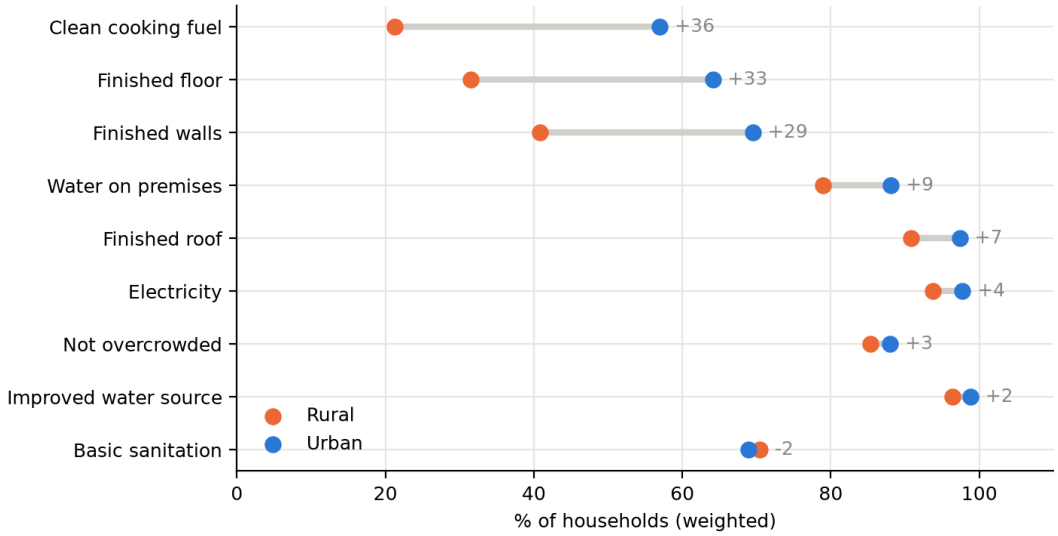



***Figure 3.*** *Each condition among urban and rural households (gap in percentage points)*

The largest differences concern clean cooking fuel and building materials. Clean fuel is used by 57.0 percent of urban households and 21.2 percent of rural households, a gap of 35.7 percentage points. Finished floors are present in 64.1 percent of urban households compared with 31.5 percent of rural households, while finished walls are present in 69.5 and 40.8 percent, respectively. The corresponding gaps are 32.6 percentage points for floors and 28.7 points for walls.

Differences in other conditions are smaller. Water is available on the premises for 88.1 percent of urban households and 78.9 percent of rural households. Finished roofs are common in both areas, although urban coverage is higher by 6.5 percentage points. The difference in households classified as not overcrowded is 2.7 percentage points.

These are unadjusted gaps. Section 5.4 presents adjusted estimates for the three indices, and Section 6.1 presents adjusted gaps for each condition.

## 5.2 Distributions of the three indices

Figure 4 shows how the individual conditions combine within households. The Basic Services Index has a similar distribution in urban and rural areas. All three basic-service conditions are met by 67.0 percent of urban households and 64.8 percent of rural households. Approximately 31 percent in each area meet two conditions. The national urban–rural distinction therefore captures relatively little variation in this index.

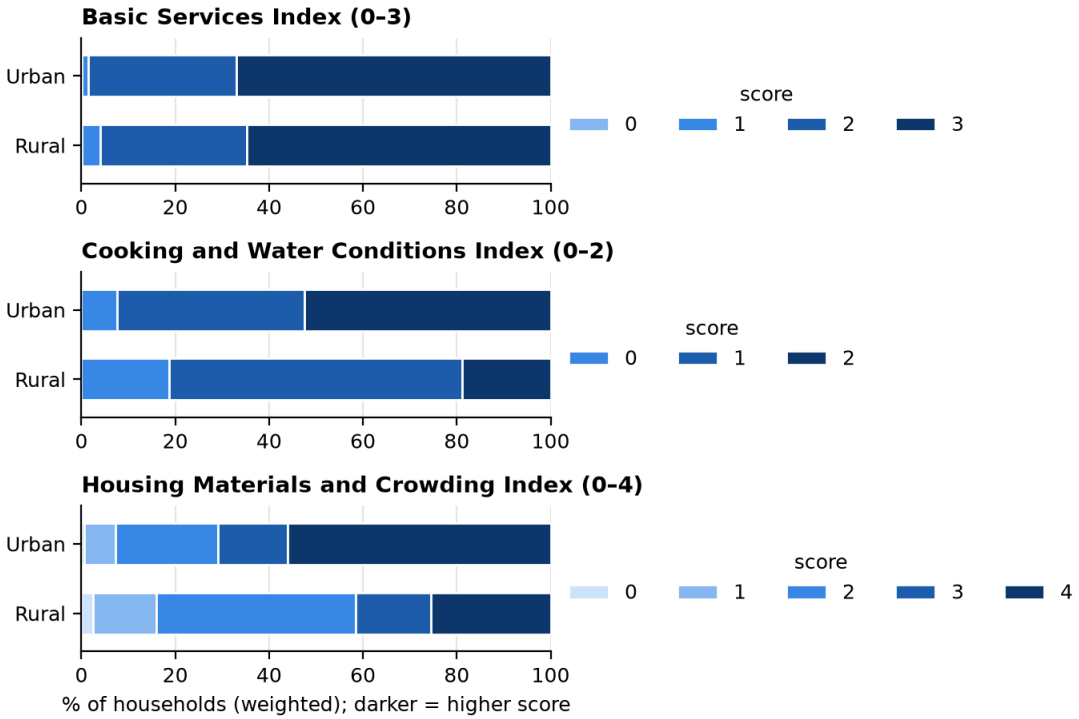



***Figure 4.*** *Distribution of index scores, urban and rural households*

The Cooking and Water Conditions Index shows a larger separation. Both conditions—clean cooking fuel and water on the premises—are met by 52.5 percent of urban households, compared with 18.8 percent of rural households. Rural households are concentrated at a score of one: 62.5 percent meet one condition, compared with 39.9 percent of urban households. Neither condition is met by 18.7 percent of rural households and 7.5 percent of urban households.

The Housing Materials and Crowding Index shows a similar urban advantage. All four conditions are met by 56.1 percent of urban households but only 25.5 percent of rural households. The most common rural score is two, accounting for 42.4 percent of rural households, compared with 22.0 percent of urban households.

## 5.3 Household profiles

Disadvantages occur together more often than the separate rates imply. Figure 5 compares the observed share of households lacking several conditions with an independence benchmark, which holds each condition's observed prevalence within the group fixed and assumes that the conditions occur independently. In urban areas, 23.5 percent of households lack clean cooking fuel, a finished floor, and finished walls together, compared with 4.7 percent under the benchmark. In rural areas, the shares are 50.3 and 31.9 percent. Lacking four or more of the nine conditions shows the same pattern: 16.6 versus 5.4 percent in urban areas and 36.1 versus 29.0 percent in rural areas. Within each of the five asset groups, the observed share lacking all three conditions also exceeds the benchmark, and the cluster-bootstrap 95 percent intervals for the difference exclude zero (Appendix Table A8). For lacking four or more conditions, the difference is not distinguishable from zero in the lowest asset group. The observed share lacking all three conditions is about five times the independence benchmark in urban areas and 1.6 times the benchmark in rural areas.

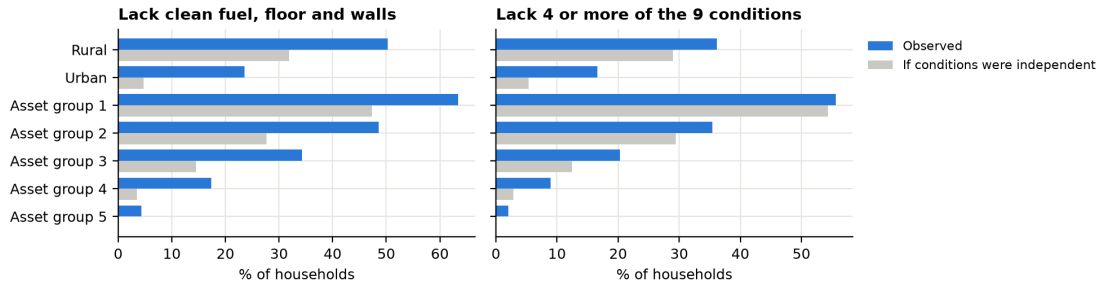



***Figure 5.*** *Observed share of households lacking several conditions and the share expected if conditions were independent within each group*

Figure 6 combines the three dimensions into the household profiles defined in Section 4. The "Basic Services and Living Conditions Met" profile requires all three basic services, both cooking and water conditions, and at least three of the four housing conditions. It includes 34.6 percent of urban households and 11.4 percent of rural households.

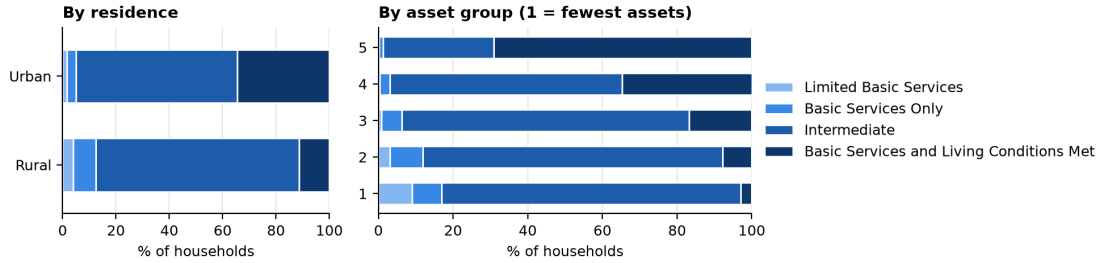



***Figure 6.*** *Household profiles by residence and asset group*

The "Basic Services Only" profile includes 8.4 percent of rural households and 3.5 percent of urban households. These households meet all three basic-service conditions but have neither clean cooking fuel nor water on the premises. The definition places no restriction on housing materials or crowding.

"Limited Basic Services," defined as meeting at most one basic-service condition, accounts for 4.0 percent of rural households and 1.5 percent of urban households. Most households fall into the Intermediate category, including 76.2 percent in rural areas and 60.4 percent in urban areas. This broad category contains several combinations of conditions and should not be interpreted as a uniform standard of living.

The share of households in "Basic Services and Living Conditions Met" rises from 2.9 percent in the lowest asset group to 69.0 percent in the highest.

## 5.4 Adjusted associations and sensitivity to the resource measure

Table 2 and Figure 7 examine urban–rural differences after adjustment for observed characteristics. The urban coefficient describes the difference in household living conditions after adjustment for the characteristics included in each model. All five specifications include province and ecological-zone controls, and the estimation sample is held constant across specifications within each outcome. Specifications 2–4 differ in the resource measure: DHS wealth quintiles, the alternative asset groups, or the continuous asset score. Specification 5 adds household-head education, sex, and age to the asset-group specification.

***Table 2.*** *Urban coefficient by specification (index points). WLS with household weights; standard errors clustered by PSU in parentheses; same households within each outcome. All specifications include province and ecological zone; (5) adds the household head's education, sex and age. * p < 0.05, ** p < 0.01, *** p < 0.001.*

| Outcome | N | (1) Geography | (2) + DHS wealth | (3) + asset groups | (4) + asset score | (5) + asset groups + head |
|---|---|---|---|---|---|---|
| BSI (0–3) | 13,757 | 0.055* (0.022) | -0.044* (0.022) | -0.019 (0.021) | -0.026 (0.021) | -0.005 (0.020) |
| CWI (0–2) | 13,714 | 0.347*** (0.036) | 0.081*** (0.024) | 0.245*** (0.031) | 0.237*** (0.031) | 0.220*** (0.030) |
| HMCI (0–4) | 13,721 | 0.528*** (0.057) | 0.080* (0.035) | 0.357*** (0.047) | 0.348*** (0.048) | 0.324*** (0.045) |

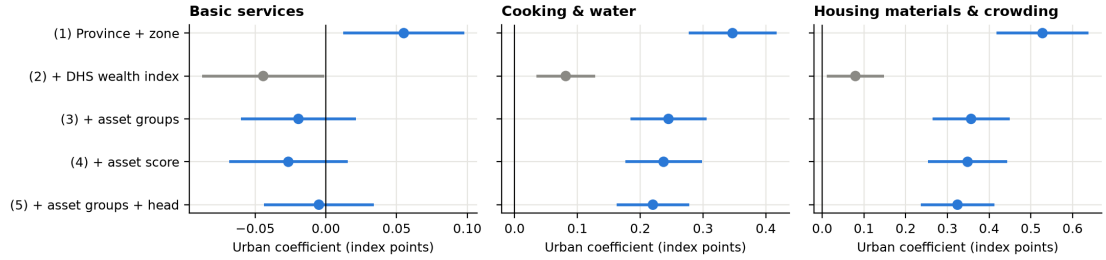



***Figure 7.*** *Urban coefficient by specification (grey = DHS wealth index; lines = 95% CI)*

For basic services, the urban coefficient is small throughout. With geographic controls alone, urban residence is associated with a BSI score higher by a small but statistically significant 0.055 points on the zero-to-three scale. With DHS wealth quintiles, the coefficient is −0.044. It becomes −0.019 with alternative asset groups and −0.005 after adding household-head characteristics; neither estimate is statistically distinguishable from zero. The results provide little evidence of an adjusted urban advantage in the combined basic-service count.

Larger associations remain for cooking and water conditions. The urban coefficient is 0.347 points with geographic controls, 0.245 with asset groups, and 0.237 with the continuous asset score. Adding household-head characteristics reduces it to 0.220 points on the zero-to-two scale. Thus, the urban association becomes smaller after adjustment but remains positive and statistically significant.

The housing-materials and crowding results follow the same pattern. The urban coefficient falls from 0.528 points with geographic controls to 0.357 with asset groups and 0.348 with the continuous asset score. With household-head characteristics included, it is 0.324 points on the zero-to-four scale. These are index-point differences, representing differences in the expected number of conditions met rather than percentage-point differences in a single outcome.

The two asset measures give similar estimates. The choice that makes a larger difference is using DHS wealth quintiles. The CWI coefficient is 0.081 with DHS wealth quintiles and 0.245 with the alternative asset groups. For HMCI, the corresponding estimates are 0.080 and 0.357. The DHS wealth index includes some of the same conditions that enter the outcomes.

I construct several versions of the asset index using the same eleven assets, households, and PCA procedure. I use each index to form asset groups and estimate the same regression (Figure 8). Adding floor, wall, and roof materials lowers the urban coefficient for housing materials and crowding from 0.357 to 0.122. Adding service and cooking indicators alone gives 0.279. With all nine outcome indicators included, the estimate falls to 0.071, close to the 0.080 obtained with DHS wealth quintiles.

In this exercise, adding the three building-material indicators accounts for most of the drop from the asset-only estimate.

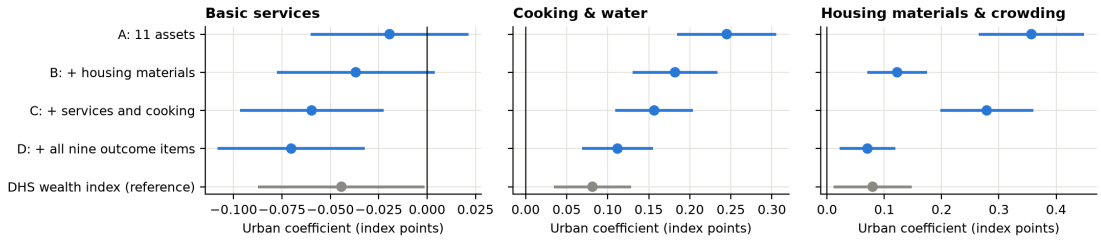



***Figure 8.*** *Urban coefficient when outcome items are added to the asset index (asset-group specification; 95% CI)*

# 6. Where the gaps are

## 6.1 Adjusted differences in individual conditions

Figure 9 examines each household condition separately using the full specification, which includes province, ecological zone, asset groups, and household-head education, sex, and age. Each binary outcome is estimated using a weighted linear probability model, with standard errors clustered by primary sampling unit. The urban coefficient, multiplied by 100, gives the adjusted difference in percentage points.

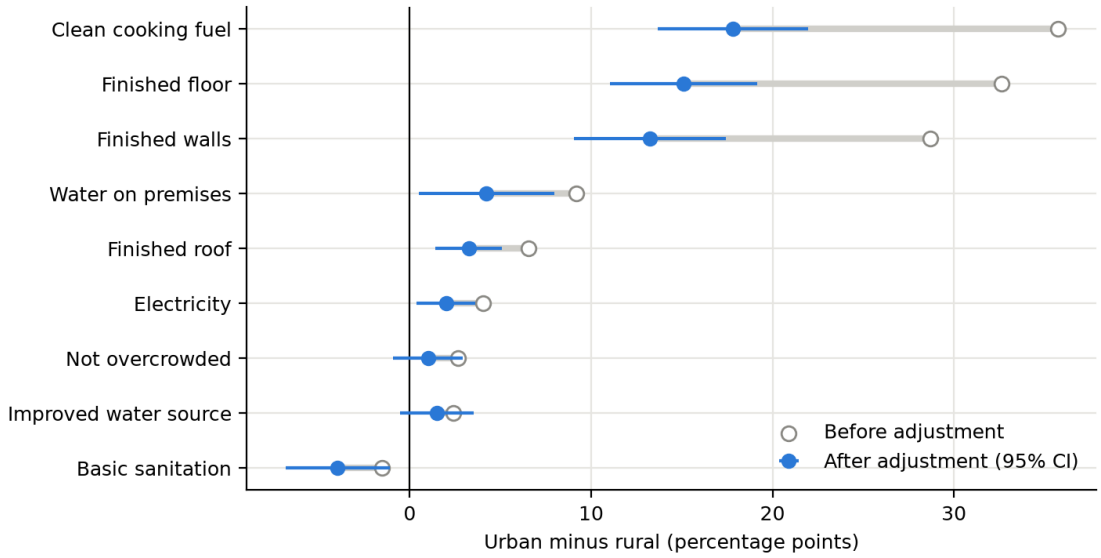



***Figure 9.*** *Urban gap in each condition, before and after adjustment (specification 5; 95% CI)*

The largest adjusted differences remain in clean cooking fuel and finished floor and wall materials. Urban residence is associated with a 17.8-percentage-point higher probability of clean-fuel use, a 15.1-point higher probability of a finished floor, and a 13.3-point higher probability of finished walls. All three associations are statistically significant at $p < 0.001$.

Each of the ten national asset groups includes at least 155 rural and 361 urban households (Appendix Table A5 and Appendix Figure A3). Some province-by-zone cells have few households from one residence group or none at all (Appendix Table A6). I allow the urban–rural gap to vary across the ten asset groups and average the group-specific estimates using the rural distribution across those groups. The resulting gaps are 16.8 percentage points for clean cooking fuel, 14.7 for finished floors, and 12.9 for finished walls, close to the main estimates of 17.8, 15.1, and 13.3 points (Appendix Table A7). Using the same households and survey weights as in specification 5 for each outcome, the unadjusted gaps are 35.7, 32.6, and 28.7 points. The main adjusted estimates are 46 to 50 percent of those unadjusted gaps.

Other adjusted urban advantages are smaller. Water on the premises differs by 4.2 percentage points, finished roofs by 3.3 points, and electricity access by 2.0 points. Each remains statistically significant at the 5 percent level. By contrast, the estimated differences in improved drinking-water sources and adequate sleeping space are 1.5 and 1.0 percentage points, respectively, and are not statistically distinguishable from zero.

Basic sanitation follows a different pattern. The rural advantage increases from 1.5 percentage points in the descriptive comparison to 4.0 points after adjustment ($p = 0.007$). Sharing is relevant to interpreting this result because shared facilities do not qualify as basic sanitation. Reported sharing is more common in urban than rural areas, at 26.6 percent compared with 22.2 percent. This descriptive difference provides context, but does not establish how much of the adjusted sanitation association is attributable to sharing.

The small adjusted BSI difference combines a 2.0-point urban advantage in electricity and a 4.0-point rural advantage in sanitation, while the estimate for improved drinking water has a confidence interval that includes zero.

## 6.2 Differences within provinces and ecological zones

Figure 10 presents household-weighted, unadjusted urban–rural differences within provinces and ecological zones. These comparisons describe geographic variation before adjustment for assets and household-head characteristics. Horizontal lines show 95 percent confidence intervals based on standard errors clustered by primary sampling unit.

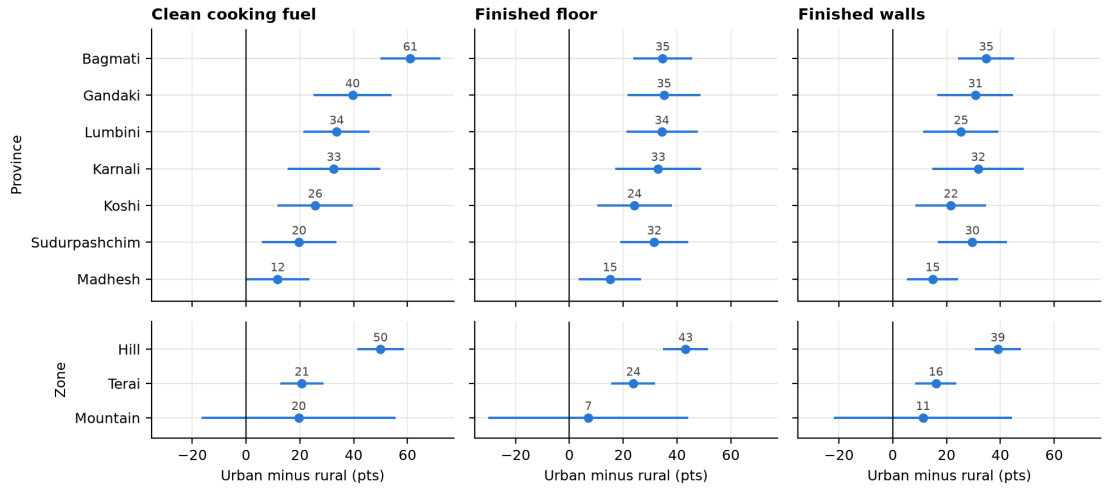



***Figure 10.*** *Urban–rural gaps by province and ecological zone (weighted, no controls; 95% CI)*

The clean-cooking gap varies considerably across provinces. Its point estimate is largest in Bagmati, at 61.0 percentage points, followed by Gandaki at 39.6, Lumbini at 33.6, and Karnali at 32.6. Smaller gaps are observed in Koshi at 25.7 points, Sudurpashchim at 19.7, and Madhesh at 11.8. The Madhesh clean-fuel gap is only marginally distinguishable from zero ($p = 0.05$). Bagmati includes the Kathmandu Valley, but this analysis does not isolate the Valley's contribution to the provincial difference.

The rankings for building materials are not identical. The finished-floor gap ranges from 15.1 percentage points in Madhesh to 35.2 points in Gandaki. The finished-wall gap ranges from 14.8 points in Madhesh to 34.7 points in Bagmati. Thus, the province with the largest estimated cooking-fuel gap does not have the largest gap in every housing indicator.

Across ecological zones, the Hill zone has the largest point estimates for all three conditions: 49.9 percentage points for clean fuel, 43.2 for finished floors, and 39.0 for finished walls. The corresponding gaps in the Terai are 20.8, 23.8, and 16.0 points. In the Mountain zone, they are 19.5, 7.0, and 11.3 points.

The Mountain estimates require particular caution. The urban Mountain sample contains only 327 households across 11 clusters, providing a much smaller basis for comparison than the provincial samples, each of which contains at least 30 urban and 30 rural clusters. The smaller Mountain point estimates, particularly for finished floors, should not be interpreted as firm evidence that urban–rural differences are smaller there. None of the three Mountain gaps is statistically distinguishable from zero, and their 95 percent confidence intervals span roughly ±37 percentage points. Regional rankings remain descriptive unless supported by direct tests of differences between the estimated gaps.

Direct tests support some of these regional differences but not others. In a regression that allows a separate urban–rural gap for each region, equality of the clean-fuel gap across provinces is rejected ($p < 0.001$), and the Bagmati gap exceeds the next largest, in Gandaki, by 21.5 percentage points ($p = 0.02$). Equality across provinces is not rejected for finished floors ($p = 0.18$) or finished walls ($p = 0.14$), so the provincial rankings for building materials are not statistically supported. Across ecological zones, equality is rejected for all three conditions ($p \leq 0.002$), and the Hill gap exceeds the Terai gap by 29.2 points for clean fuel, 19.4 for finished floors, and 23.0 for finished walls ($p \leq 0.001$ in each case).

## 6.3 Interpretation

Clean fuel, finished floors, and finished walls account for the largest adjusted differences. Before adjustment, the clean-fuel gap varies across provinces, and the Hill zone has larger gaps than the Terai on all three measures. The analysis leaves open how much these differences reflect fuel prices and availability, construction markets, or resources the asset measure misses.

# 7. Robustness

Figure 11 compares the linear estimates under different weights, asset scores, material definitions, and geographic samples. Figure 12 reports the ordered-logit estimates. Unless stated otherwise, the checks use specification 5, with province, ecological zone, asset groups, and household-head education, sex, and age.

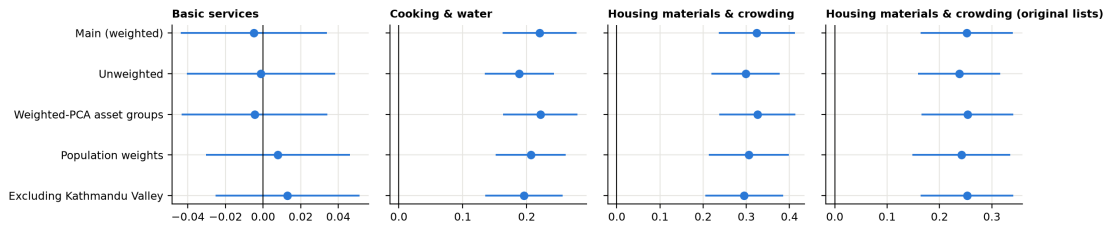



***Figure 11.*** *Urban coefficient across robustness checks (95% CI)*

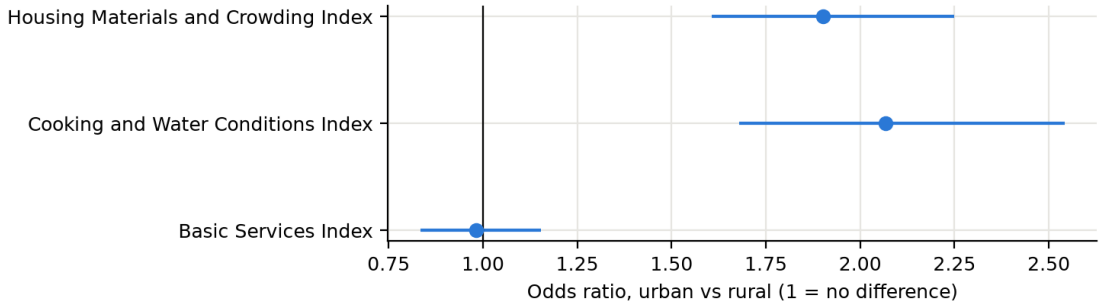



***Figure 12.*** *Ordered logit: odds ratio of a higher score, urban vs rural (95% CI)*

## 7.1 Weighting, asset-index construction, and survey design

Removing survey weights produces somewhat smaller urban coefficients for cooking and water conditions and for housing materials and crowding. The CWI coefficient falls from 0.220 to 0.188, while the HMCI coefficient falls from 0.324 to about 0.30. Both remain positive and statistically significant at $p < 0.001$. The BSI estimate remains close to zero, changing from −0.005 to −0.001.

Building the asset groups with survey-weighted PCA changes the urban coefficients by at most 0.002 index points. The resulting estimates are −0.005 for BSI, 0.221 for CWI, and 0.326 for HMCI.

I also multiply the household weights by the number of usual residents, giving more weight to conditions in larger households. The estimated urban coefficients are 0.008 for BSI, 0.206 for CWI, and 0.305 for HMCI. The BSI estimate remains statistically indistinguishable from zero.

I also compute standard errors by Taylor linearization using the household weights, primary sampling units, and 14 strata. The design-based standard errors are close to the PSU-clustered estimates in all six fully adjusted models (Appendix Table A9). For clean cooking fuel, the standard error is 2.06 percentage points with stratification and 2.11 with PSU clustering alone. The two approaches give the same conclusions about statistical significance for the six estimates shown.

## 7.2 Material definitions and Kathmandu Valley

The housing result is more sensitive in magnitude to the classification of finished materials. Using an alternative classification of finished materials reduces the HMCI urban coefficient from 0.324 to 0.252 points, with $p < 0.001$. These lists count wood-plank floors as finished and exclude carpet, wood-plank or shingle walls, and some roof materials included in the main definition. The estimated magnitude therefore depends on the classification, but the positive urban association remains.

Excluding Kathmandu, Lalitpur, and Bhaktapur districts tests whether the findings persist outside Kathmandu Valley. These districts account for 812 households in 28 clusters, including 784 urban

households, and represent 15.8 percent of weighted urban households. After their exclusion, the urban coefficient is 0.196 for CWI and 0.295 for HMCI, compared with 0.220 and 0.324 in the full sample. Both remain statistically significant at $p < 0.001$. The BSI coefficient is 0.013 and is not statistically significant. Thus, the urban associations in cooking and housing conditions extend beyond the capital region.

### 7.3 Ordered-logit specification

The indices count conditions met, and a linear model treats each additional condition as an equal increment in the outcome. Ordered logit provides a supplementary specification that treats the scores as ordered categories. These models use the same control set and PSU-clustered standard errors, but are unweighted because the implementation used here does not support survey weights.

Urban residence is associated with higher categories of CWI and HMCI. The urban odds ratio is 2.07 for CWI, with a 95 percent confidence interval of 1.68–2.54, and 1.90 for HMCI, with an interval of 1.61–2.25. For BSI, the odds ratio is 0.98, with an interval of 0.83–1.15. These results are consistent with the qualitative pattern in the linear models. The odds ratios describe cumulative odds of higher scores under the ordered-logit model; they are not directly comparable with index-point coefficients.

Across the linear checks, BSI estimates stay close to zero, and the CWI and HMCI estimates remain positive. The alternative materials list makes the largest difference to HMCI, lowering the estimate from 0.324 to 0.252 points.

## 8. Which households would different targeting approaches reach?

I compare four illustrative targeting rules using the household survey weights. The main comparison focuses on households that lack clean cooking fuel, a finished floor, and finished walls together. These are the three conditions with the largest urban–rural gaps, and 32.4 percent of households lack all three. I also examine households lacking four or more of the nine conditions (23.0 percent). I report the share of households each rule selects, the share of households meeting each definition of need that the rule selects, and the share of selected households that lack all three conditions. Confidence intervals use 300 bootstrap resamples clustered by primary sampling unit.

Rural households are more likely to lack all three conditions, but urban municipalities contain about two-thirds of Nepal's households. As a result, 48.7 percent of households lacking all three live in urban municipalities (95 percent confidence interval, 42.7 to 54.6). The urban share among households lacking four or more conditions is 48.2 percent. Koshi and Madhesh account for the largest shares of households lacking all three conditions, at 24.2 and 18.9 percent, respectively. In Madhesh, about two-thirds of these households live in urban municipalities (Appendix Table A11).

Figure 13 compares the four rules. The rural rule selects 33.0 percent of households and reaches 51.3 percent of those lacking all three conditions. The basic-service rule selects 33.6 percent and reaches 41.3 percent of households lacking all three, but reaches 70.2 percent of households lacking four or more conditions. The bottom two groups of the separate asset index select 38.5 percent of households and reach 64.6 percent of those lacking all three conditions. The bottom two DHS wealth quintiles select 40.3 percent and reach 89.2 percent. Among households selected, the share lacking all three conditions is 50.3 percent under the rural rule, 39.7 percent under the basic-service rule, 54.3 percent

under the separate asset rule, and 71.7 percent under the DHS wealth rule. Appendix Table A10 gives confidence intervals.

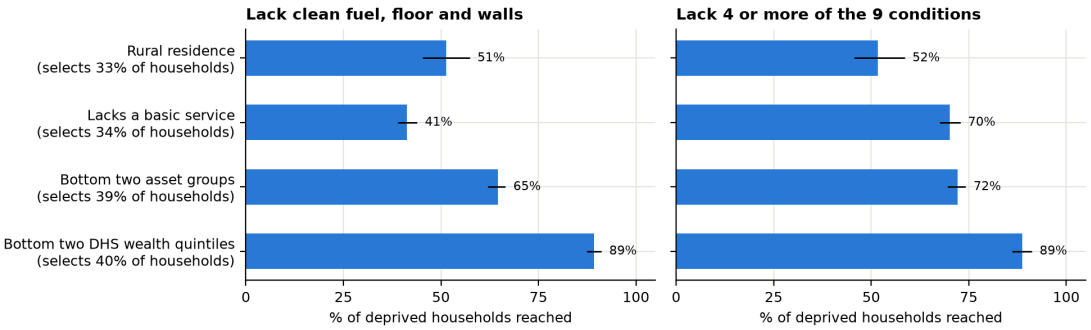



***Figure 13.*** *Share of deprived households each targeting rule would reach (95% CI from 300 PSU-cluster bootstrap resamples)*

The DHS wealth index includes some of the conditions used to define need, which helps explain its high reach in this sample. In Section 5.4, that overlap also changes the interpretation of the adjusted urban–rural gap. The rules select different shares of households, and the exercise measures whom each rule would flag in the survey. Actual program reach would also depend on eligibility, outreach, and take-up.

# 9. Conclusion, implications and limitations

## 9.1 Main findings

Urban households lead by 36 percentage points in clean cooking fuel, 33 in finished floors, and 29 in finished walls. Those gaps fall to 18, 15, and 13 points after adjustment for geography, assets, and household-head characteristics. The adjusted basic-service count differs little by residence. Electricity and improved water are widespread, while roughly three in ten households fall short of basic sanitation in both areas. The estimated clean-fuel gap is largest in Bagmati, and the Hill zone has larger gaps than the Terai in clean fuel, finished floors, and finished walls. The weighting and specification checks preserve the main pattern.

## 9.2 Implications for measurement

The estimated urban difference in housing materials and crowding is 0.53 index points with geographic controls alone. Adding DHS wealth quintiles lowers it to 0.08. Adding the separate asset groups instead gives 0.36, close to the result using the continuous asset score.

The DHS wealth control partly reflects the same housing and service conditions the regressions are trying to explain. Reporting estimates with both the DHS wealth index and a resource measure built from separate items shows how much this choice changes the comparison.

Adding the nine outcome indicators to the asset index lowers the estimated housing gap from 0.36 to 0.07 index points. The result shows how much the adjusted gap can change when the wealth control includes the conditions being studied.

The basic-service count also shows the value of reporting individual indicators. The small adjusted difference in the combined score reflects, in part, an urban advantage in electricity and a rural advantage in basic sanitation.

## 9.3 Policy implications

Programs addressing these cooking and housing disadvantages should include urban municipalities in outreach and check the specific conditions they aim to improve. Deprivation rates show where the conditions are most common. The distribution of households facing them helps determine where more outreach may be needed. The targeting results describe the households each rule would identify in the 2022 survey.

Monitoring electricity and improved water alone would miss the largest gaps in this study. Reporting cooking fuel and building materials separately would show where differences persist.

About 21 percent of rural households use a clean main cooking fuel. The survey identifies the size of the gap. Assessing responses would require evidence on fuel prices, availability, infrastructure, and household budgets.

About three in ten households in both urban and rural areas lack basic sanitation. Shared facilities are more common in urban areas and fall short of the basic sanitation definition even when the facility type is improved. Reporting facility type and sharing separately would show which condition limits coverage.

The larger Hill–Terai gaps and the variation in clean fuel across provinces identify places for closer study. The provincial tests give weaker evidence of differences in the floor and wall gaps.

## 9.4 Limitations

First, the survey covers one period, so the estimates describe differences in 2022 and cannot identify causal effects or changes over time. The urban classification follows municipality status, which can group settlements with very different densities.

Second, the indices are equally weighted counts, and the household profiles use selected thresholds. Neither provides a comprehensive measure of welfare. The underlying indicators also capture particular characteristics rather than complete service or housing quality. An improved drinking-water source does not establish safety or reliability, and finished building materials do not establish structural safety.

Third, the asset index is an imperfect proxy for household resources. Asset ownership reflects infrastructure, geography, and preferences as well as purchasing power, and the score is correlated with urban residence. The regressions adjust for the observed asset groups or score, depending on the specification. The asset score is not a direct measure of income or consumption.

Fourth, the main standard errors account for clustering by primary sampling unit. Including the 14 sampling strata gives similar standard errors for the six models checked in Section 7.1. Regional comparisons are unadjusted, and the Mountain-zone estimates rely on only 11 urban clusters. Their precision and comparability therefore require particular caution.

Finally, the migration measure varies at the district level and records absentee prevalence rather than household remittance receipt or amounts. The absence of statistically significant associations in that supplementary analysis does not demonstrate the absence of a household-level relationship.

## 9.5 Future work

Household-level remittance and housing data, potentially from NLSS-IV when suitable microdata are accessible, would allow a more direct examination of the remittance relationship. Earlier NDHS rounds (2001–2016) would allow a comparison across repeated cross-sections, showing how quickly coverage has risen and whether urban–rural gaps in clean cooking fuel and building materials have narrowed. Such comparisons require harmonizing indicator definitions across questionnaires and accounting for the 2017 reclassification of urban areas, for example by comparing ecological zones or districts rather than administrative urban status alone. A density-based classification would also help assess how sensitive the findings are to the administrative definition of urban residence.

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# Appendix A. Supplementary tables and figures

***Appendix Table A1.*** *Classification of every response category (1 = condition met, 0 = not met, missing = excluded). Households are unweighted counts.*

| Indicator | DHS variable | Response category | Score | Households |
|---|---|---|---|---|
| water | hv201 | piped to yard/plot | 1 | 5,585 |
| water | hv201 | tube well or borehole | 1 | 4,033 |
| water | hv201 | public tap/standpipe | 1 | 1,648 |
| water | hv201 | piped into dwelling | 1 | 889 |
| water | hv201 | bottled water | 1 | 637 |
| water | hv201 | piped to neighbor | 1 | 283 |
| water | hv201 | protected spring | 1 | 272 |
| water | hv201 | unprotected spring | 0 | 194 |
| water | hv201 | protected well | 1 | 102 |
| water | hv201 | unprotected well | 0 | 87 |
| water | hv201 | river/dam/lake/ponds/stream/canal/irrigation channel | 0 | 46 |
| water | hv201 | tanker truck | 1 | 9 |
| water | hv201 | other | missing | 1 |
| toilet | hv205 | flush to pit latrine | 1 | 6,053 |
| toilet | hv205 | flush to septic tank | 1 | 5,000 |
| toilet | hv205 | no facility/bush/field | 0 | 910 |
| toilet | hv205 | flush to piped sewer system | 1 | 655 |
| toilet | hv205 | pit latrine with slab | 1 | 464 |
| toilet | hv205 | biogas attached toilet | 1 | 446 |
| toilet | hv205 | ventilated improved pit latrine (vip) | 1 | 148 |
| toilet | hv205 | pit latrine without slab/open pit | 0 | 58 |
| toilet | hv205 | flush to somewhere else | 0 | 28 |
| toilet | hv205 | flush, don’t know where | 1 | 11 |
| toilet | hv205 | other | missing | 6 |
| toilet | hv205 | composting toilet with slab | 1 | 5 |
| toilet | hv205 | bucket toilet | 0 | 1 |
| toilet | hv205 | composting toilet without slab | 0 | 1 |
| fuel | hv226 | wood | 0 | 8,260 |
| fuel | hv226 | liquefied petroleum gas | 1 | 4,548 |

| Indicator | DHS variable | Response category | Score | Households |
|---|---|---|---|---|
| | | (lpg)/cooking gas | | |
| fuel | hv226 | animal dung/waste | 0 | 356 |
| fuel | hv226 | straw/shrubs/grass | 0 | 220 |
| fuel | hv226 | biogas | 1 | 179 |
| fuel | hv226 | agricultural crop | 0 | 77 |
| fuel | hv226 | electricity | 1 | 54 |
| fuel | hv226 | no food cooked in house | missing | 40 |
| fuel | hv226 | solar energy | 1 | 15 |
| fuel | hv226 | charcoal | 0 | 15 |
| fuel | hv226 | coal/lignite | 0 | 6 |
| fuel | hv226 | piped natural gas | 1 | 4 |
| fuel | hv226 | gasoline/diesel | 0 | 4 |
| fuel | hv226 | sawdust | 0 | 4 |
| fuel | hv226 | processed biomass (pellets) or woodchips | 0 | 3 |
| fuel | hv226 | kerosene/paraffin | 0 | 1 |
| floor | hv213 | cement | 1 | 4,794 |
| floor | hv213 | earth/sand | 0 | 4,570 |
| floor | hv213 | dung | 0 | 3,103 |
| floor | hv213 | carpet | 1 | 719 |
| floor | hv213 | vinyl or asphalt strips | 1 | 292 |
| floor | hv213 | ceramic tiles | 1 | 130 |
| floor | hv213 | wood planks | 0 | 103 |
| floor | hv213 | parquet or polished wood | 1 | 65 |
| floor | hv213 | other | missing | 8 |
| floor | hv213 | palm/bamboo | 0 | 2 |
| wall | hv214 | cement | 1 | 5,164 |
| wall | hv214 | stone with mud | 0 | 3,258 |
| wall | hv214 | mud/sand | 0 | 1,860 |
| wall | hv214 | bamboo with mud | 0 | 1,299 |
| wall | hv214 | bricks | 1 | 1,267 |
| wall | hv214 | metal/galvanized sheet | 0 | 225 |
| wall | hv214 | wood planks/shingles | 1 | 173 |
| wall | hv214 | stone with lime/cement | 1 | 159 |
| wall | hv214 | cement blocks | 1 | 153 |
| wall | hv214 | cane/palm/trunks | 0 | 136 |
| wall | hv214 | reused wood | 0 | 43 |
| wall | hv214 | plywood | 0 | 23 |
| wall | hv214 | other | missing | 11 |
| wall | hv214 | cardboard | 0 | 9 |
| wall | hv214 | no walls | 0 | 6 |
| roof | hv215 | metal/galvanized sheet | 1 | 6,344 |
| roof | hv215 | cement | 1 | 3,485 |
| roof | hv215 | roofing shingles | 1 | 1,651 |
| roof | hv215 | ceramic tiles | 1 | 1,012 |
| roof | hv215 | thatch/palm leaf | 0 | 583 |
| roof | hv215 | calamine/cement fiber | 1 | 339 |

| Indicator | DHS variable | Response category | Score | Households |
|---|---|---|---|---|
| roof | hv215 | mud | 0 | 283 |
| roof | hv215 | wood planks | 0 | 42 |
| roof | hv215 | other | missing | 22 |
| roof | hv215 | palm/bamboo | 0 | 11 |
| roof | hv215 | rustic mat | 0 | 5 |
| roof | hv215 | wood | 1 | 5 |
| roof | hv215 | no roof | 0 | 4 |

***Appendix Table A2.*** *Asset items and their loadings on the first principal component. No household is missing any item.*

| Item | DHS variable | Weighted % owning | Loading | Loading (weighted PCA) |
|---|---|---|---|---|
| radio | hv207 | 19.8 | 0.10 | 0.12 |
| television | hv208 | 48.7 | 0.39 | 0.38 |
| refrigerator | hv209 | 26.6 | 0.43 | 0.43 |
| bicycle | hv210 | 36.9 | 0.14 | 0.07 |
| motorcycle | hv211 | 27.9 | 0.41 | 0.40 |
| car/truck/tractor | hv212 | 5.0 | 0.23 | 0.23 |
| landline | hv221 | 3.8 | 0.27 | 0.28 |
| mobile phone | hv243a | 95.8 | 0.18 | 0.18 |
| watch | hv243b | 57.0 | 0.28 | 0.30 |
| computer | hv243e | 13.6 | 0.38 | 0.38 |
| bank account | hv247 | 78.6 | 0.28 | 0.28 |

***Appendix Table A3.*** *Urban–rural gap in each condition, percentage points. Adjusted = linear probability model, specification 5, PSU-clustered standard errors (Figure 9).*

| Condition | Raw gap | Adjusted gap | SE | p |
|---|---|---|---|---|
| Clean cooking fuel | 35.7 | 17.8 | 2.1 | 0.000 |
| Finished floor | 32.6 | 15.1 | 2.1 | 0.000 |
| Finished walls | 28.7 | 13.3 | 2.1 | 0.000 |
| Water on premises | 9.2 | 4.2 | 1.9 | 0.027 |
| Finished roof | 6.5 | 3.3 | 0.9 | 0.001 |
| Electricity | 4.0 | 2.0 | 0.8 | 0.017 |
| Not overcrowded | 2.7 | 1.0 | 1.0 | 0.302 |
| Improved water source | 2.4 | 1.5 | 1.0 | 0.151 |
| Basic sanitation | -1.5 | -4.0 | 1.5 | 0.007 |

***Appendix Table A4.*** *Reconciliation with the NDHS 2022 report. "People" weights each household by its usual residents. Report values: electricity 98 (urban) and 94 (rural); clean cooking 52.6 (urban, Table 2.4); basic sanitation 72.6 (all, Table 16.7).*

| Indicator | Households, urban | People, urban | People, rural | People, all |
|---|---|---|---|---|
| Electricity | 97.7 | 97.8 | 94.0 | 96.5 |
| Clean cooking fuel | 57.0 | 52.6 | 19.3 | 41.5 |
| Basic sanitation | 68.9 | 72.7 | 72.6 | 72.7 |

***Appendix Table A5.*** *Asset distribution by residence: ten asset groups from the weighted asset-score distribution, with tied scores kept together (so group sizes differ). Households are unweighted counts.*

| Asset group | Rural, weighted % | Urban, weighted % | Rural households | Urban households |
|---|---|---|---|---|
| 1 | 13.8 | 7.3 | 970 | 643 |
| 2 | 7.2 | 4.5 | 483 | 361 |
| 3 | 21.5 | 11.9 | 1,424 | 993 |
| 4 | 10.5 | 7.8 | 761 | 600 |
| 5 | 11.8 | 8.5 | 783 | 667 |
| 6 | 10.7 | 10.5 | 679 | 741 |
| 7 | 9.7 | 11.7 | 604 | 832 |
| 8 | 6.7 | 11.9 | 417 | 793 |
| 9 | 5.5 | 11.8 | 315 | 747 |
| 10 | 2.6 | 14.1 | 155 | 818 |

***Appendix Table A6.*** *Sample households by province, ecological zone and residence (unweighted).*

| Province | Ecological zone | Rural households | Urban households |
|---|---|---|---|
| Bagmati | Hill | 721 | 1,048 |
| Bagmati | Mountain | 261 | 61 |
| Bagmati | Terai | 31 | 144 |
| Gandaki | Hill | 779 | 791 |
| Gandaki | Mountain | 28 | 0 |
| Gandaki | Terai | 90 | 117 |
| Karnali | Hill | 580 | 751 |
| Karnali | Mountain | 304 | 118 |
| Koshi | Hill | 502 | 291 |
| Koshi | Mountain | 114 | 58 |
| Koshi | Terai | 445 | 744 |
| Lumbini | Hill | 428 | 172 |
| Lumbini | Terai | 551 | 843 |
| Madhesh | Terai | 909 | 1,135 |
| Sudurpashchim | Hill | 413 | 217 |
| Sudurpashchim | Mountain | 231 | 90 |
| Sudurpashchim | Terai | 204 | 615 |

***Appendix Table A7.*** *Urban gap under a flexible asset adjustment. Flexible = urban × ten asset groups plus geography and household head, with the urban gap averaged over rural households' asset groups. PSU-clustered standard errors.*

| Outcome | Specification 5 | SE | Flexible, rural asset distribution | SE |
|---|---|---|---|---|
| BSI (index points) | -0.005 | 0.020 | -0.012 | 0.021 |
| CWI (index points) | 0.220 | 0.030 | 0.209 | 0.029 |
| HMCI (index points) | 0.324 | 0.045 | 0.320 | 0.045 |
| Clean cooking fuel (pts) | 17.8 | 2.1 | 16.8 | 2.0 |
| Finished floor (pts) | 15.1 | 2.1 | 14.7 | 2.0 |

| Outcome | Specification 5 | SE | Flexible, rural asset distribution | SE |
|---|---|---|---|---|
| Finished walls (pts) | 13.3 | 2.1 | 12.9 | 2.1 |

***Appendix Table A8.*** *Households lacking several conditions: observed share and the share expected if conditions were independent within the group (weighted %). Difference = observed minus expected, percentage points; 95% intervals from 300 PSU-cluster bootstrap resamples.*

| Group | Lack fuel, floor and walls: observed | If independent | Difference, 95% CI | Lack 4+ of 9: observed | If independent | Difference, 95% CI |
|---|---|---|---|---|---|---|
| Rural | 50.3 | 31.9 | [17.0, 19.8] | 36.1 | 28.9 | [5.2, 9.0] |
| Urban | 23.5 | 4.7 | [16.8, 20.8] | 16.6 | 5.4 | [10.1, 12.2] |
| Asset group 1 | 63.4 | 47.3 | [14.0, 18.2] | 55.6 | 54.4 | [-0.8, 3.2] |
| Asset group 2 | 48.6 | 27.6 | [19.3, 22.6] | 35.4 | 29.4 | [4.8, 7.3] |
| Asset group 3 | 34.3 | 14.5 | [18.6, 21.3] | 20.3 | 12.4 | [6.7, 8.9] |
| Asset group 4 | 17.3 | 3.5 | [12.1, 15.6] | 8.9 | 2.8 | [5.0, 7.3] |
| Asset group 5 | 4.3 | 0.1 | [3.3, 5.4] | 2.1 | 0.1 | [1.3, 2.8] |

***Appendix Table A9.*** *Standard errors with and without the sampling strata (specification 5). Design-based errors use Taylor linearization with weights, PSUs and the 14 strata.*

| Outcome | Urban coefficient | SE, PSU-clustered | SE, design-based with strata |
|---|---|---|---|
| BSI (index points) | -0.005 | 0.020 | 0.020 |
| CWI (index points) | 0.220 | 0.030 | 0.030 |
| HMCI (index points) | 0.324 | 0.045 | 0.045 |
| Clean cooking fuel (pts) | 17.82 | 2.11 | 2.06 |
| Finished floor (pts) | 15.10 | 2.07 | 2.08 |
| Finished walls (pts) | 13.25 | 2.14 | 2.15 |

***Appendix Table A10.*** *Targeting rules (weighted %, 95% intervals from 300 PSU-cluster bootstrap resamples). Urban share of households lacking all three: 48.7 [42.7, 54.6]; of households lacking four or more: 48.2 [41.5, 54.3].*

| Targeting rule | Households selected | Reaches: lack fuel, floor and walls | Reaches: lack 4+ of 9 | Selected who lack all three |
|---|---|---|---|---|
| Rural residence | 33.0 [29.1, 37.9] | 51.3 [45.4, 57.3] | 51.8 [45.7, 58.5] | 50.3 [46.7, 54.0] |
| Lacks a basic service | 33.6 [32.0, 35.3] | 41.3 [39.0, 43.9] | 70.2 [67.6, 72.8] | 39.7 [36.0, 44.0] |
| Bottom two asset groups | 38.5 [36.4, 41.0] | 64.6 [62.0, 66.5] | 72.2 [69.5, 74.2] | 54.2 [50.9, 57.4] |
| Bottom two DHS wealth quintiles | 40.2 [36.9, 44.1] | 89.2 [87.3, 91.1] | 88.8 [86.1, 91.2] | 71.7 [69.3, 74.5] |

***Appendix Table A11.*** *Where households lacking clean fuel, a finished floor and finished walls live (weighted % of all such households).*

| Province | Rural | Urban | Total |
|---|---|---|---|
| Koshi | 11.4 | 12.8 | 24.2 |
| Madhesh | 6.5 | 12.3 | 18.9 |
| Lumbini | 9.6 | 5.0 | 14.6 |

| Province | Rural | Urban | Total |
|---|---|---|---|
| Sudurpashchim | 6.9 | 7.1 | 13.9 |
| Karnali | 6.8 | 5.2 | 12.0 |
| Bagmati | 5.5 | 2.9 | 8.4 |
| Gandaki | 4.6 | 3.5 | 8.1 |

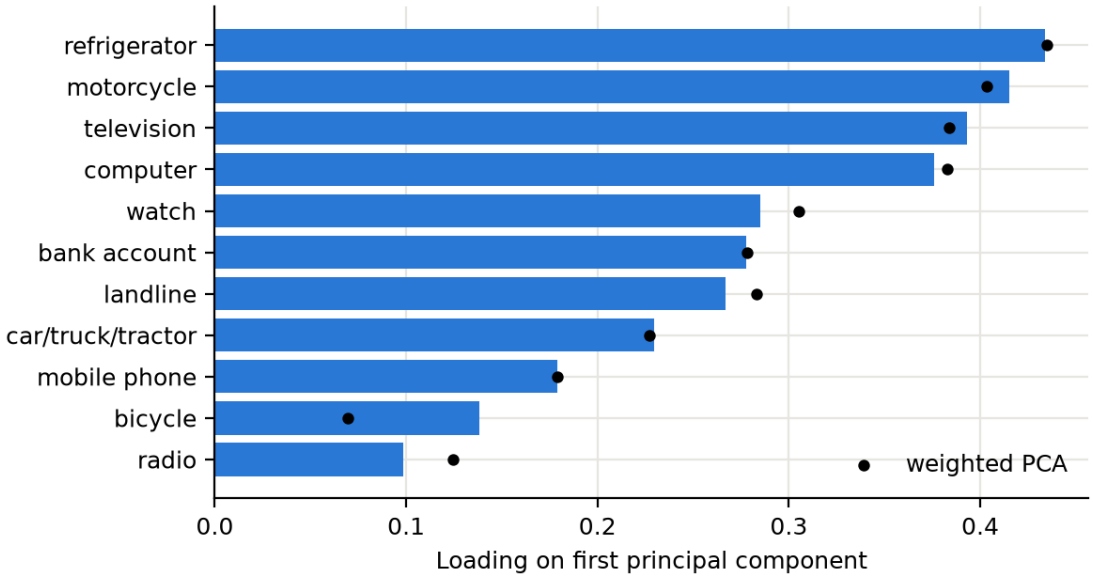



***Figure A1.*** *Asset index: loading of each item on the first principal component*

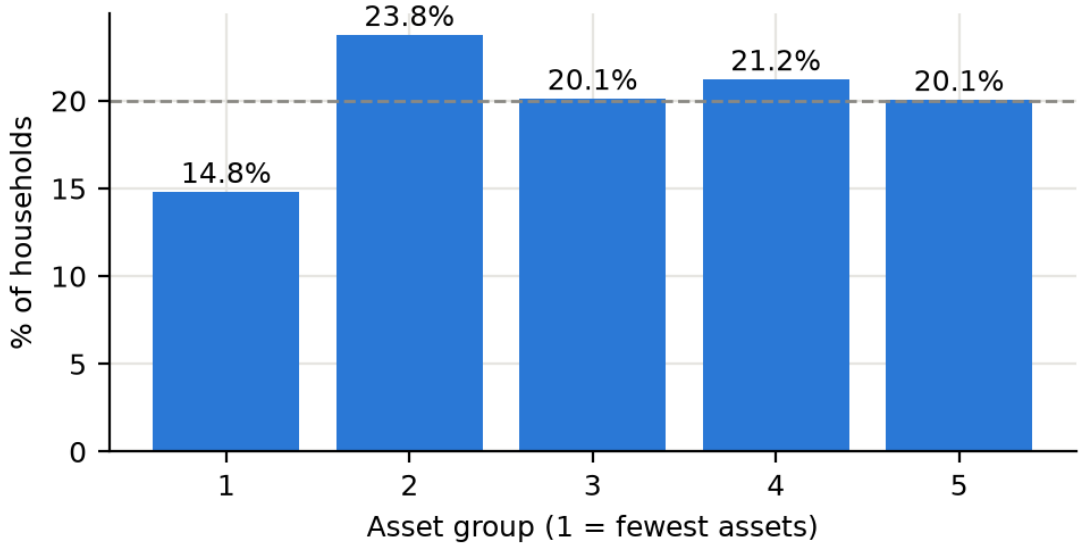



***Figure A2.*** *Asset group sizes (dashed line = exact quintile)*

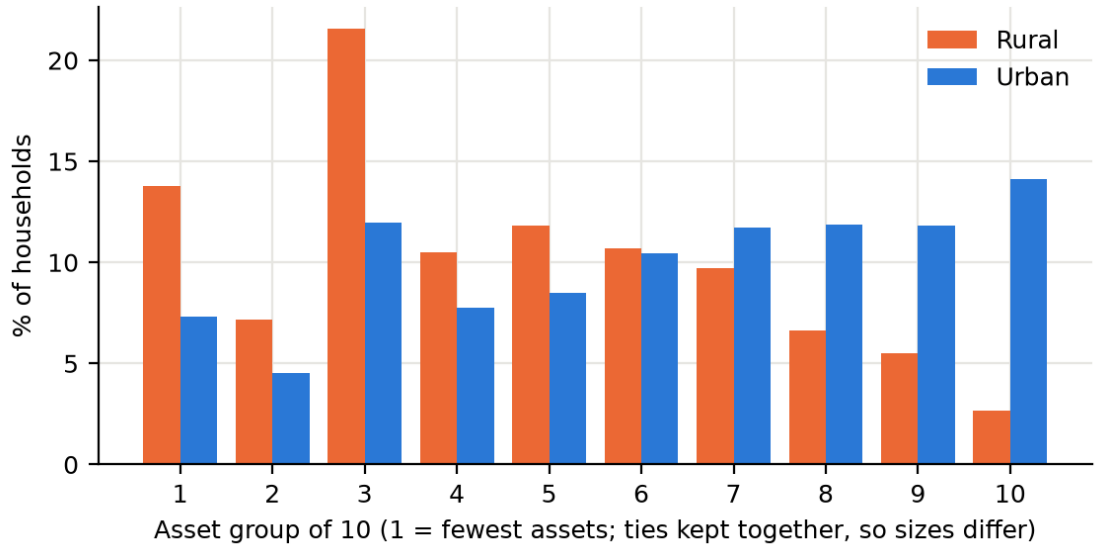



***Figure A3.*** *Asset distribution by residence: ten tie-preserving asset groups (weighted %)*

# Appendix B. District migration and household living conditions

## B.1 Question and measure

This supplementary analysis examines whether household living conditions and urban–rural differences are associated with district migration prevalence. Each household is assigned its district's Census 2021 count of absentees abroad per 100 native residents, as described in Section 3.5. All 75 districts represented in the DHS are matched. The rate ranges from approximately 1.0 in Dolpa to 19.5 in Achham; its median is 6.9 among the 73 districts containing rural DHS households.

The measure captures the prevalence of migration across districts. A district with a high absentee rate can contain many households that receive no remittances.

## B.2 Descriptive relationships

Figure B1 plots absentee rates against rural housing and cooking conditions in the 73 districts with rural observations. The unweighted correlations are 0.14 for mean rural HMCI and 0.02 for rural clean-fuel use.

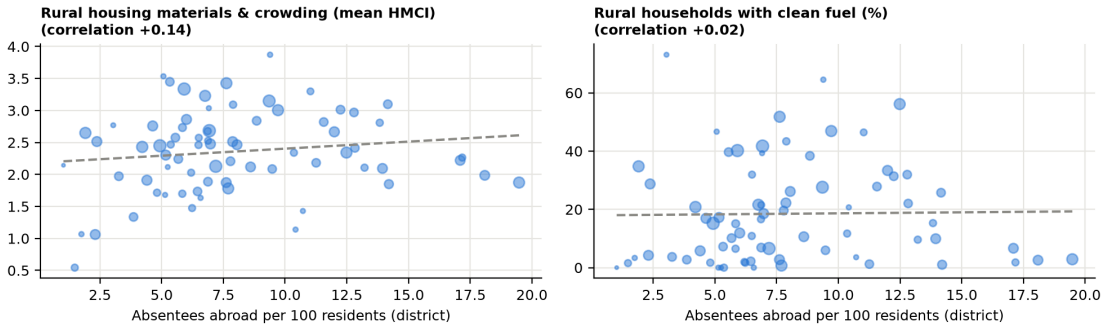



***Figure B1.*** *Rural living conditions by district migration (dot size = households)*

Each district receives equal weight in these correlations, regardless of its population or DHS sample size. They therefore describe relationships between district summaries and differ from the household-weighted regression estimates below.

## B.3 Regression results

The regression specification includes urban residence, district migration prevalence, their interaction, and household controls:

$$Y_{id} = \alpha + \beta Urban_{id} + \gamma Migration_d + \delta(Urban_{id} \times Migration_d) + X'_{id}\theta + \varepsilon_{id}.$$

Migration is scaled so that one unit represents five additional absentees abroad per 100 native residents. The coefficient $\gamma$ describes the association among rural households, while $\delta$ describes how the urban–rural difference varies with migration prevalence. Models use household sampling weights and standard errors clustered by district, giving 75 clusters.

Two control sets are considered: province and ecological zone alone, followed by the addition of asset groups and household-head characteristics. Neither the rural migration coefficients nor the interaction coefficients are statistically significant at the 5 percent level under either specification.

With the full controls, the rural migration coefficient is 0.021 index points for BSI, −0.004 for CWI, and 0.035 for HMCI. Their respective 95 percent confidence intervals are −0.030 to 0.072, −0.073 to 0.064, and −0.102 to 0.171. The estimates therefore do not identify a clear positive or negative association.

The component results are similarly inconclusive. Per five additional absentees abroad per 100 native residents, the estimated rural association is −0.8 percentage points for clean cooking fuel, −0.1 points for finished floors, and −3.3 points for finished walls. The corresponding confidence intervals are −6.7 to 5.2, −5.9 to 5.8, and −9.8 to 3.1 percentage points.

There is also no clear evidence that migration prevalence is associated with a narrower urban–rural difference. Under the full controls, the interaction estimate is −2.4 percentage points for clean cooking fuel, with a confidence interval of −8.1 to 3.3. The interactions for finished floors and walls are positive, but their intervals also include zero. Figure B2 presents the estimates and their uncertainty.

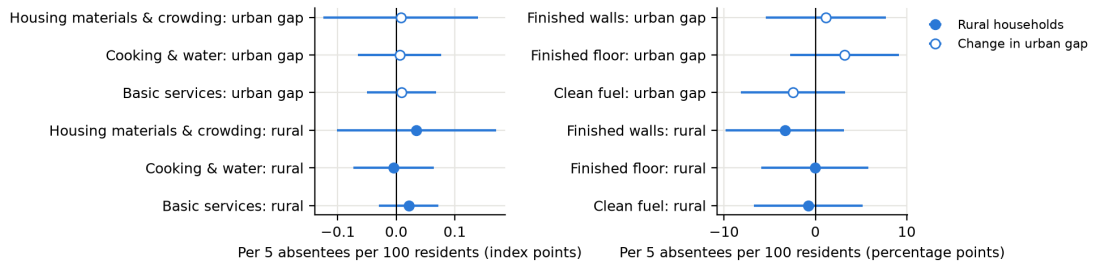



***Figure B2.*** *District migration and living conditions (asset and head controls; 95% CI)*

## B.4 Interpretation and limitations

The estimates are imprecise enough to leave room for associations of practical size. For example, the full-control interval for rural clean-fuel use ranges from a 6.7-point decrease to a 5.2-point increase per five additional absentees abroad per 100 native residents.

Geographic concentration limits the variation available for estimation. Province indicators account for approximately 58 percent of the between-district variation in absentee rates in a district-level regression. Including province controls therefore leaves the migration coefficients identified from variation within provinces, together with the other controls. Moreover, migrant and non-migrant households in the same district receive the same migration value, concealing differences in household exposure.

A direct analysis of remittances would require household-level measures of receipt or amounts together with housing and cooking conditions. The district absentee rate cannot distinguish between households that receive transfers and their neighbors.